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An Empirical Evaluation of Ground Effect for Small-Scale Rotorcraft*

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***Abstract*— Ground effect refers to the apparent increase in lift that an aircraft experiences when it flies close to the ground. For helicopters, this effect has been modeled since the 1950's based on the work of Cheeseman and Bennett, perhaps the most common method for predicting hover performance due to ground effect. This model, however, is based on assumptions that do not hold for small-scale rotorcraft because it was developed specifically for conventional helicopters. It is not clear if the Cheeseman-Bennett model can be applied to today's multirotor UAVs. In this paper, we compare the Cheeseman-Bennett model to experimental results for rotor performance due to ground effect in several small-scale multirotor and single-rotor configurations. Experimental findings suggest that some of the conventional thinking surrounding helicopter ground effect cannot be applied directly to rotorcraft using fixed propellers at variable speeds (e.g. multirotors), and that it is necessary to adjust the helicopter models to better reflect the differences in such aircraft. The experimental results for multirotors presented are for multiple propeller configurations, speeds and spacings. Ultimately, this work will facilitate the development of an improved UAV flight controller that can accurately account for ground effect to improve flight stability near surfaces and structures.**

I. BACKGROUND & RELATED WORK

The underlying mathematics that describe propeller function and thrust generation dates back centuries. Modern propeller theory can be traced back to 1865 with the work of Rankine [18], [17], and [16], who recognized that a propeller generates thrust as a reaction to accelerating a mass of air in the opposite direction (Newton's 3rd Law of Motion). Eventually, this led to what is now known as *Momentum Theory*, and is one of the most widely accepted methods for describing propellers. Momentum Theory is based on the assumptions that a stationary propeller in a moving air stream can be represented as an infinitely thin disk of diameter D across which the pressure increases discontinuously, and the stream velocity far ahead of the propeller (V_o) and far behind the propeller ($V_o + \omega$) is uniform, this is shown in Fig. 1. Additionally, the static pressure is assumed to be constant and equal to the free-stream pressure. It is important to recognize that under these conditions, the cross-section of the flow in the ultimate wake of the propeller is smaller than the disk area of the propeller. This theory is computationally simple for estimating basic quantities such as thrust (T), power (P), efficiency, and the propeller induced velocity (ω) as shown by equations (1), (2), and (3). It is clear that the maximum thrust occurs when the stream velocity $V_o = 0$ (static thrust), which is the assumed case for (3).

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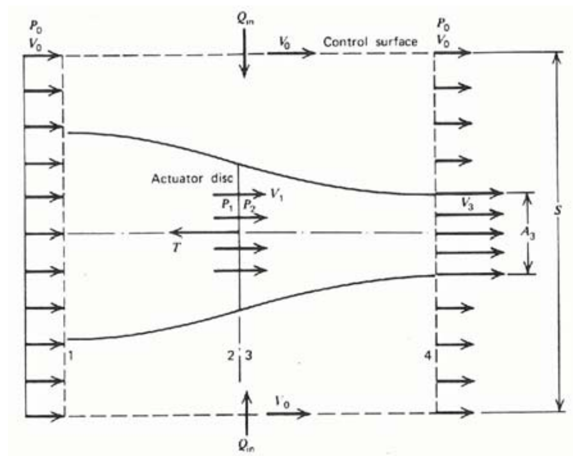

Fig. 1: Idealized flow diagram for classical Momentum Theory. [16].

Momentum Theory offers computational simplicity for basic propeller analysis; however, it does not take into account the actual propeller geometry (pitch, airfoil, number of blades, etc.), nor the fact that the propeller induces both angular and axial velocity components to the air stream. To address these limitations, Stefan Drzewiecki published a theory in 1909 based on the opposite approach of a rotating blade element advancing axially through stationary air [17] and [16]. This so-called *Blade Element Theory* often combines pieces of Momentum Theory to provide a more complete description of propeller performance, sometimes called *Blade Element Momentum Theory*.

In the same manner as the linear motion of a fixed-wing, the rotating airfoil of a propeller blade generates aerodynamic lift and drag, which ultimately determine the thrust and required torque. More recently however, it became evident that the vortices observed in the wake of a moving airfoil were the result of bound circulation around the airfoil, and directly related to the production of lift. Likewise, this bound circulation occurs around the blades of a propeller and thus, directly relates to thrust. *Vortex Theory* [16] and [19] quantifies this bound circulation (Γ) and ultimately provides a numerical method for directly calculating the induced velocity (ω) in terms of axial (ω_a) and tangential (ω_t) components using (4) as demonstrated by [20], where α_i is the induced angle of attack. Combined, these three theories form a complete mathematical description of propellers.

$$T = 2\rho A \omega (V_o + \omega) \quad (1)$$

$$P = 2\rho A \omega (V_o + \omega)^2 \quad (2)$$

$$\omega = \sqrt{\frac{T}{2\rho A}} = \frac{P}{T} \quad (3)$$

$$\omega_t = V_R \alpha_i \sin(\phi + \alpha_i) \quad and \quad \omega_a = V_R \alpha_i \cos(\phi + \alpha_i) \quad (4)$$

It is well known that aircraft experience significant aerodynamic interactions when flying close to the ground. Ground-effect manifests as an apparent increase in lift experienced as the aircraft approaches the ground. Ground effect for helicopters can be described as the increase in main rotor lift per unit power (this ratio is called power loading) experienced when flying close to the ground. The increase in power loading is inversely proportional to the aircraft's height above the ground, and is generally insignificant at heights greater than twice the effective rotor diameter. Ultimately, ground effect is the result of a reduction in the induced velocity from the rotor when flying close to the ground, as illustrated in Fig. 2. When high above the ground, the induced velocity is higher because the downward airflow is unobstructed. Close to the ground, the downward flow is slowed and forced to disperse radially as it encounters the ground. According to Blade Element Momentum Theory, thrust is inversely proportional to induced velocity, explaining the apparent increase in thrust in ground effect.

Conventional helicopter ground effect has been well studied since the 1950s, with the most widely accepted mathematical model published by Cheeseman and Bennett in 1957 [1]. The Cheeseman-Bennett model [1] is most commonly defined by the equation in the form of (5) which provides a relationship between the radius (R) of the propeller, the height above the ground (Z), the thrust generated by the rotor while operating far from the ground (T_{OGE}), and the apparent increase in thrust experienced close to the ground (T_{IGE}). However, this model assumes a single variable-pitch rotor turning at a constant speed using constant engine power.

These assumptions are not valid for multirotor aircraft because they have multiple variable-speed, fixed-pitch rotors. Additionally, conventional helicopter rotor blades typically have a uniform airfoil profile along the span; whereas, multirotors typically use tapered airfoils with a decreasing cord length and angle of attack across the span from root to tip. Additional variables such as changing rotor speeds, number of rotors, rotor placement (e.g. different spacing and angles between rotor supports), dihedral, and possibly ducted/shrouded rotors can all apply to multirotors and their influence on ground effect is not well studied. A few attempts have been made to comprehensively model ground-effect in multirotors, but do not address the overarching assumptions used to simplify the problem [7], [9], and [10]. For example, (5) predicts that the ground effect becomes insignificant at heights greater than 4R (two rotor diameters), however [7] experimentally demonstrates that this is not necessarily accurate for small-scale multirotors. The results presented

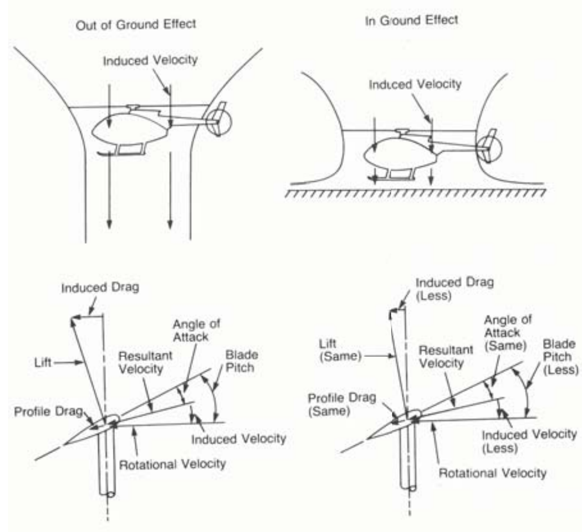

Fig. 2: Ground effect according to Blade Element Theory [11].

here demonstrate that (5) cannot be applied to multirotors because it does not accurately predict the change in lift due to ground effect.

Similar effects occur when flying close to ceilings, walls, and other large barriers / obstacles. Rotorcraft, in particular, are susceptible to these effects, especially while hovering or performing delicate low-speed maneuvers [11]. As with fixed-wing aircraft, rotorcraft can exhibit significant variations in control sensitivity and responsiveness during takeoff and landing because of ground effect. Any roll or pitch corrections near the ground can result in unequal and disproportionate loading of the rotor disk due to one side being closer to the ground than the opposite side. In other words, ground effect can create instability by changing the aircraft's response to a given control input. Additionally, maintaining a constant rate of ascent / decent is more difficult near the ground because the power loading is variable. Many UAV flight controllers ignore these problems, treating ground effect like any other disturbance and intentionally minimizing the time spent in immediate proximity to the ground during takeoff and landing. It is desirable to develop an improved flight controller that includes model-based prediction of ground effect, as well as near-wall and near-ceiling effects. This work seeks to study and quantify ground effect in a comprehensive manner for multirotor aircraft in an effort to develop this improved flight controller in the future.

$$\frac{T_{IGE}}{T_{OGE}} = \frac{1}{1 - \left(\frac{R}{4Z}\right)^2} \qquad iff \qquad \frac{Z}{R} > 0.25 \quad (5)$$

The Cheeseman-Bennett model employs the single equation and boundary condition in (5) to conveniently calculate the ratio of the lift experienced while hovering in-ground-effect (IGE) and out-of-ground-effect (OGE). Because the equation only requires two basic measurements to predict the ground effect, it has been widely utilized. Other studies of the era [2], [3] validate the Cheeseman-Bennet equation (5) using experimental data to focus on a change in thrust approach. This equation assumes that the main rotor speed and power output both remain constant during IGE and OGE flight. These assumptions may be reasonable for helicopters, but they are inherently impossible for multirotors that are continuously varying the rotor speeds to generate control authority.

In the two decades after [1] was published, researchers had already noticed several factors that could influence the ground effect predictions. The work published in [4] and [5] present evidence that bench tests using an isolated rotor could not recreate the ground effect performance increase that was seen in full scale flight tests. A flying aircraft benefits more from ground effect due to the decreased download on the airframe/fuselage that is directly beneath the rotor. A bench test however, does not experience this benefit, and thus the results will produce lower T_{IGE} values under similar conditions. Equation (5) also does not account for the lift loading or the number of rotor blades. This means that a smaller propeller with greater pitch is capable of producing the same thrust under the same conditions as a larger propeller, but (5) cannot account for this. Therefore, the equation can overestimate T_{IGE} for a large rotor with low lift loading, and underestimate T_{IGE} for a small rotor with high lift loading.

The gradual shift from conventional helicopter UAVs to multirotor UAVs, has led some researchers to study ground effect on multirotor platforms using (5) from the Cheeseman-Bennett model. In the work of [6], the ground effect on a typical quadrotor is considered using (5) for the purpose of modeling and control development, a goal that is shared by many UAV researchers. The work presented in [7] goes into greater detail using experimental data and theory to describe the effect on a quadrotor of flying close to the ground, as well as vertical walls. Similarly, the work in [10] uses computational fluid dynamics (CFD) to model the aerodynamic interaction between a hovering helicopter UAV and a vertical wall. This work provides valuable insight into visualizing the airflow and its interaction behavior with a surface.

Building upon [6], [7] and [8], the work presented in [9] uses (5) to evaluate the ground effect on a Draganflyer X8 octorotor that features four coaxial pairs of propellers. This work treats each coaxial pair as a single rotor based on Blade Element Momentum Theory, therefore equating the aircraft to a quadrotor. The claim that the ground effect for a quadrotor can modeled by (5) as four single rotors summed together is shown to be inaccurate by the results presented below. This work also uses a test stand to isolate one coaxial pair of rotors and evaluate them before conducting flight tests. Because only a single (coaxial pair) rotor is evaluated, the authors fail to address how possible interactions between multiple flow streams may affect ground effect performance. This work does not directly measure T_{IGE}, but rather assumes it to be equal to the aircraft takeoff weight. Additionally, this work does not directly measure the rotor power, but rather assumes that the power at a given RPM is constant in both OGE and IGE cases.

It is clear that (5) from the Cheeseman-Bennett model has been widely used, however it is based on assumptions that are only valid for conventional helicopters that use large rotors with uniform blades and variable pitch spinning at a constant speed. However, the assumptions are not valid for rotorcraft with fixed-pitch multipurpose propellers that spin at variable speeds. Additionally, these propellers typically have a tapered airfoil that varies both the angle of attack and the cord length across the span of the blade. These fundamental differences in aircraft design suggest a possible disconnect when attempting to apply conventional helicopter ground effect models to another rotorcraft such as a multirotor. Classical theory indicates these variables may not have any effect, however, this research investigates all possibilities that may influence ground effect rotor performance.

$$\Delta HP = \frac{T v_{1OGE}}{550}\left[1-\left(\frac{\omega_{IGE}}{\omega_{IGE}}\right)\right] \tag{6}$$

$$P_{IGE} = P_{OGE}\left(\frac{\omega_{IGE}}{\omega_{IGE}}\right) \tag{7}$$

An alternative model presented in [11] describes the decrease in power consumption seen in the IGE case using (6), rather than the increase in thrust. Equation (6) is for power measured in HP units only, but it can be reduced to the dimensionless ratio form in (7). Assuming that the power required during OGE hover (P_{OGE}) can be measured directly, the power required during IGE hover (P_{IGE}) can be calculated using equation (7) . The induced velocities at constant thrust ω_{IGE} and ω_{OGE} can be calculated using an equation such as (3), experimentally determined, or from results of previous published experiments such as [11], [12] and [13].

This model is equally useful in predicting ground effect performance, and it is not limited by assuming constant power and speed. Instead, this model assumes a constant thrust condition in both OGE and IGE cases. This assumption is easy to apply to any electric powered rotorcraft, because the aircraft weight is equal to the overall downward thrust during hover. Because electric aircraft do not consume fuel during flight, their weight remains constant. Therefore, the constant thrust assumption holds true during hover if the minor thrust variations due to attitude corrections average out over any non-zero length of time.

This paper aims to demonstrate that ground effect for small-scale multirotor aircraft cannot be accurately described using conventional ground effect theories such as (5). Multiple factors can have a dramatic effect on hover performance near the ground, and in many cases this effect is contrary to the predictions made by the existing conventional models.

II. METHODS & PROCEDURES

This paper focuses on the experimental evaluation of multiple quadrotor configurations to characterize IGE and OGE hover performance due to ground effect. The experimental apparatus allows for incremental adjustment of three fundamental parameters: 1) the distance between the propeller plane and the ground (Z), 2) the spacing between the propellers on a typical X-shaped quadrotor frame, and 3) the speed of the motors. The experimental setup is focused on eliminating uncontrolled and undesirable influences as much as possible to isolate only the desired test parameters. Every effort is made to minimize mechanical vibration, EMI, air flow obstruction, temperature variations, etc.

The experiments are performed indoors for climate control purposes in a large loading bay with a 5 m ceiling that does not influence ground effect measurements. The experimental setup is inverted so that the airflow directed upward against a large movable flat surface that serves as the "ground" plane. There are two reasons for this: 1) the test frame is at a fixed height so a movable "ground" surface is required, and 2) the ceiling is sufficiently high so that an upward airflow had more space to disperse without obstruction. The ground height incrementation is accomplished using a large structure that supports a 2.4 m square sheet of reinforced plywood that recreates a ground surface. The vertical posts of the structure have holes at increments of 2.5 cm, through which metal rods are inserted to support the plywood surface. The height is incremented by lifting each side of the plywood slightly, and reinserting the metal rod into the next hole on the structure. While simplistic, this apparatus makes it possible to evaluate ground effect in controlled small increments through a large range of heights from ≈3-120 cm.

Based on preliminary research, a single motor and propeller combination is chosen that is representative of the most common small/medium scale quadrotors that are widely used by researchers and hobbyists. The 150 W Leopard LC2830-980kV motors are comparable in size and power to the most common quadrotor motors offered by numerous manufacturers. The Gemfan 9 x 4.7 nylon propellers are flexible, inexpensive, widely available and generally more popular compared to composite propellers. The Castle Creations Phoenix Edge 50A Electronic Speed Controllers (ESCs) are not typical for quadrotor on this scale, but their high-quality construction and special features (speed feedback, data logging, superior programmability, etc.) are desirable for this specialized experimentation.

A. Instrumented Test Stand

A custom fabricated test stand, shown in Fig. 3, that minimizes airflow obstruction and mechanical vibration is constructed to be sufficiently far from ceilings, floors, and walls so as not to influence the measurements. The test stand features welded steel construction and a central hollow tube with removable end-caps, and is filled with 20 kg of sand to further reduce vibration. An ATI Mini-40 6-DOF force / torque sensor is rigidly mounted to the top of the test stand, to which the aircraft test frame is directly attached. This sensor measures the reaction forces and torques experienced by the aircraft test frame in response to the test parameters. This sensor has a capacity of 80 N (F_x, F_y), 240N (F_z), and 4 Nm (T_x, T_y, T_z), with a maximum resolution of 0.02 N and 0.0005 N-m. These measurements along with the motor speeds are recorded through a National Instruments (NI) USB 6002 DAQ which is interfaced with a computer via USB 2.0.

Fig. 3: Custom instrumented test stand with the ATI Mini40 F/T sensor.

Fig. 4: Custom fabricated quadrotor test frame composed of 92 cm thick-wall square tubing bolted to a $\frac{1}{4}$ inch plate. Multiple motor mounting locations are located at 50 mm increments along each arm.

A custom fabricated X-shaped quadrotor test frame, shown in Fig. 4 is specifically designed for this experimental application with heavy-duty construction using thick-wall aluminum square tubing to maximize rigidity and heat dissipation from the motors. The test frame is machined with multiple motor mounting locations at 50 mm increments along each arm for exploring how the propeller spacing distance influences T_{IGE}, T_{OGE}, and thrust-power efficiency. Additionally, a different experimental setup is used to evaluate an isolated single propeller under the same test conditions as a point of comparison.

The motors / ESCs require a 12 VDC power source capable of producing at least 60 A overall. Initial attempts using various battery power sources resulted in measurement inconsistencies due to the voltage drop under load and as the state-of-charge decreased. This is normal for all batteries, however it was determined to be an unacceptable drawback

Fig. 5: Experimental setup for the single isolated propeller case using the motor / propeller combination throughout this paper.

Fig. 6: Two Astec DS550-3 power supplies connected in parallel using active load sharing.

because a constant motor speed could not be maintained for more than a few seconds. Instead, an AC-DC power supply unit (PSU) is used, consisting of two parallel Astec (Artesyn) DS550-3 distributed network power supplies for a total output of 90 A at 12 VDC, shown in Fig. 6. The DS550-3 has an active load sharing feature that allows multiple units to be connected in parallel without the need for external rectifiers. Heavy 6AWG cables carry the DC power to the ESCs and a 33,000 uF capacitor minimizes transient voltage fluctuations under changing load conditions.

B. Testing Procedures

The raw data from the F/T sensor is inherently noisy due to mechanical vibrations from the motor and propeller. An averaging filter is implemented because of its computational simplicity and overall effectiveness. The tests are all performed using a sampling rate of 1 kHz, and an averaging size of 250 samples. This results in an effective sampling rate of 40 Hz, with each sample consisting of 6 component measurements. The motor speeds and supply voltage are measured separately without filtering. Additional data recorded by the ESCs, including voltage, current, power, RPM, throttle input, temperature, and ripple voltage is downloaded from each ESC at the conclusion of each experiment.

Incrementing the speed of the motors directly affects thrust, disk loading, power loading, and induced velocities. Because the power output of the motor is directly related to propeller speed, an increase in speed also represents an increase in power, however the relationship is not linear. Therefore, when the propeller speed is constant, it can be assumed that, ideally, the motor power is also constant. A closed-loop speed controller is implemented by matching the timing of the motor speed feedback signals from the ESCs. A master-slave approach modulates the PWM signals for three of the ESCs to match the speed of the master ESC receiving the PWM signal set by the operator. This method ensures the motor speeds are always identical. While this is unlikely for a real quadrotor in flight responding to disturbances and unequal payload distribution, it is essential for replicating the ideal static hover scenario that is desirable for controlled empirical evaluation.

The data collection procedure begins with a 3 minute warm-up cycle to get the motors to their typical operating temperature of ≈40°C. Motor temperature has a significant impact on performance, so it is critical to maintain a consistent operating temperature by carefully timing the experimental duty cycle. At each increment of the test parameters described above, the motors are held at a constant throttle for a 10 second data collection. After each 10 second test, the throttle is reduced to a minimum level for 90 seconds to avoid heat buildup in the motors. During the 90 second dead-time the test parameter is incremented again, after which the process repeats. This procedure is used throughout all of the experiments, and the temperature and humidity is always recorded.

In total, 396 separate data collections are recorded and analyzed in this study, exploring numerous combinations of the test parameters mentioned above. Three different motor speeds are tested at 22 height increments from the ground surface between ≈3-120 cm. In addition to the isolated single propeller case, data was gathered for five different propeller spacings on the quadrotor test frame, 39-79 cm diagonally in 10 cm increments.

III. RESULTS

This section presents the measured thrust performance of the chosen motor / propeller combination spinning at constant speed as a function of the height from the ground surface. Thrust performance is represented by the dimensionless ratio T_{IGE}/T_{OGE} in all cases, where 1 is the performance level when the propeller(s) is very far from the ground (OGE). When the ratio is greater than unity, there is an increase in thrust due to the ground effect. For comparison, the theoretical predicted thrust performance is calculated from (5) and represented by a dashed line. The comparison between measured and predicted thrust performance is discussed in detail.

A. Single Isolated Propeller

The classical theories of propeller analysis were developed for an isolated single-propeller case. Likewise, the conventional theories for helicopter ground effect in (5) and (7) that only consider a single main rotor. Beginning with the evaluation of an isolated single propeller serves to validate the experimental setup, and forms a basis for comparison

Fig. 7: Single motor and propeller in operation during testing.

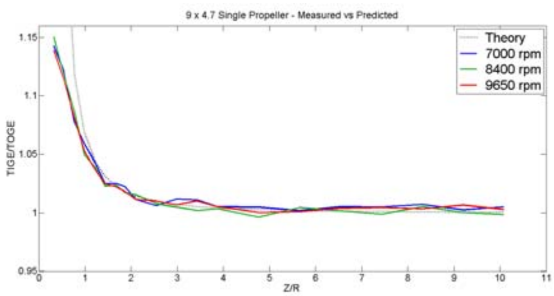

Fig. 8: Thrust performance curves due to ground effect for a single isolated propeller.

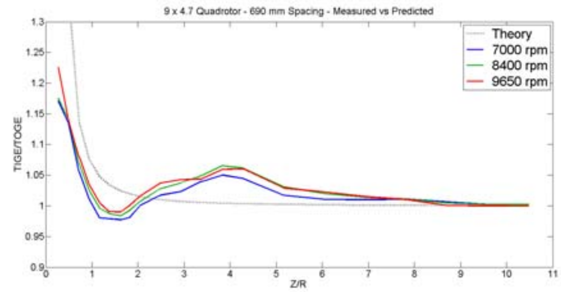

Fig. 9: Thrust performance curves due to ground effect for 690 mm quadrotor configuration.

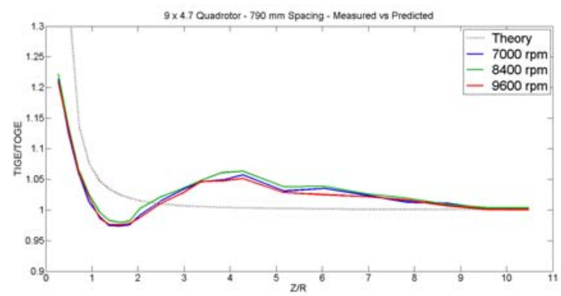

Fig. 10: Thrust performance curves due to ground effect for 790 mm quadrotor configuration.

to identify how the interactions between multiple propellers affect the overall hover performance. The data shown in Fig. 8 describes the measured and predicted thrust performance of a single motor and propeller at constant speed. Fig. 7 shows this experimental setup during operation. This plot clearly shows the significant increase in thrust performance very close to the ground, and because this is a dimensionless value, it is essentially the independent of propeller speed and/or motor power. The plot also illustrates the rapid decrease in thrust performance as height increases, decaying asymptotically to 1, and ground effect becomes insignificant above $Z \simeq 4R$. All of these observations are characteristic of the classical ground effect model defined by (5), and can be seen in both the measured and predicted performance curves in all cases. The curves do not match up perfectly however, which could be attributed to the fact that small-scale aircraft operate in a sensitive range of Reynolds number that is not true of large scale aircraft. In the small scale, complex flow patterns can exist in the boundary region around the airfoil of the propeller which can have a significant effect on the thrust performance [21].

B. Quadrotor Thrust Performance

The presumption that the thrust performance of a multirotor can be treated as the sum of single rotors is an oversimplification of a multifaceted problem. By applying (5) to a single rotor, and multiplying by the number of rotors (n), as suggested in [7] and [9] is equivalent to having one propeller of the same diameter that produces n times the thrust. According to Momentum Theory, the cross-section of the accelerated flow in the ultimate wake of the propeller is smaller than the disk area of the propeller, shown in Fig. 1. Therefore, without obstruction, the multiple discrete flow streams produced by a multirotor will not combine into one larger flow, but rather become more differentiated as they continue far beyond the propellers along parallel axes.

Close to the ground, it is necessary to consider how these flow streams will interact when they encounter the ground surface. As each discrete flow encounters the surface, it cannot spread out radially in all directions due to the adjacent flows. Colliding air from adjacent flow streams hitting the surface prevents the even radial spread seen in Fig. 2, further limiting the induced velocity of each flow. It is logical to expect an increase in thrust performance, as a result, that would not occur in a single-rotor situation.

The plots in Figs. 9 and 10 show several thrust performance curves for various quadrotor configurations. It is immediately evident that the measured thrust performance differs significantly from the predictions made by (5) as well as the single-propeller performance curves. As discussed, there is a significant increase in thrust performance that does not exist in single rotor cases, and as a result, is not predicted by (5). It is interesting to note that this increase does not occur closest to the ground, but rather at a distance around $Z \simeq 4R$ where ground effect is supposed to become insignificant according to (5) and the single-propeller measurements.

Equally notable, between $Z \simeq 1.5R$ and $Z \simeq 2R$, the ground surface actually produces a detrimental effect on thrust performance, which briefly drops below 1. This can be explained by considering the collision of discrete air flows at the ground surface described above. Since the airflow cannot uniformly spread out radially where it collides with an adjacent flow, some of the colliding air will be forced to curl away from the surface and away from the colliding flow. This sets up a poloidal flow condition that is characteristic of a vortex ring, which have been observed in similar situations such as microbursts. When a vortex ring recirculates around a spinning rotor, it can result in a loss of thrust that cannot be corrected by increasing power, known in aviation as a *vortex ring state*. Figs. 9 and 10 support this line of reasoning, since the reduced thrust performance in this region exists in curves measured at different power levels. When sufficiently far from the ground surface, any vortex rings produced are too far downstream to recirculate through the propellers and there is no loss of thrust.

IV. CONCLUSIONS & FUTURE WORK

This paper aims to determine if the classical Cheeseman-Bennett model of ground effect in full-scale conventional helicopters also applies to small-scale propellers and multirotors. Through the application of mathematical principles from Momentum Theory and Blade Element Theory, combined with empirical evaluation and analysis, the research presented here suggests that this model cannot accurately predict ground effect performance for quadrotors using (5). This conclusion is further validated by demonstrating the accuracy of the experimental process through evaluation of a single isolated propeller that more closely parallels a conventional helicopter, and can be accurately described by (5). Further experimentation using a broader set of test parameters, including different propeller sizes and number of rotors (hexarotor and octorotor configurations), will provide the basis for developing a new set of mathematics that can accurately predict multirotor ground effect. Future studies using this approach will be evaluate near-wall effect as well. This comprehensive study and modeling of ground effect that is specific to multirotors will eventually lead to more advanced flight control development that can apply model-predictive approaches to flying in ground effect and near-wall effect regions.

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